

Scale Specific Riemannian Features with Graph Convolutional Network for Motor Imagery EEG Classification

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Abstract—Motor imagery (MI) electroencephalogram (EEG) signals, generated during imagined limb movements, provide essential neural patterns for brain–computer interface (BCI) applications. Accurate decoding of these signals is critical for robust MI-BCI systems. This work proposes a unified framework integrating multi-scale wavelet decomposition, Riemannian geometry, and graph-based deep learning for MI-EEG classification. Continuous Wavelet Transform with Morlet wavelets extracts scale-specific time–frequency features, from which spatial covariance matrices are computed. Fisher Geodesic Discriminant Analysis enhances class separability on the Riemannian manifold, and the features are projected into tangent space before concatenation across scales. A Graph Convolutional Network then models trial-level relational structure, treating each trial as a graph node connected via k-nearest neighbors. Comparative evaluation with Support Vector Machine, Linear Discriminant Analysis, and Artificial Neural Network baselines shows superior performance. On the BCI Competition IV-2a dataset, the proposed method achieves 96.36% (subject-dependent) and 80.94% (subject-independent), surpassing prior approaches by 13.23% and 12.81%, respectively.

Index Terms—Electroencephalogram, Motor Imagery, Continuous Wavelet Transform, Riemannian manifold, Graph Convolutional Network

I. INTRODUCTION

Motor imagery (MI)-based brain–computer interfaces (BCIs) have emerged as a promising pathway for restoring communication and motor functions in individuals with severe neuromuscular impairments [1], [2]. Such systems hold promise for assistive technologies, neurorehabilitation, and control of external devices. Among available neuroimaging techniques, electroencephalography (EEG) remains the most widely used due to its high temporal resolution, portability, and

non-invasive nature as well as relatively low cost. The central challenge in this paradigm lies in accurately decoding the subtle spatio–temporal dynamics of EEG signals to distinguish between imagined movements, such as left- or right-hand motor imagery [3]. To advance research in this area, the BCI Competition IV-2a [4] dataset has become a benchmark resource. It provides multi-class MI recordings from multiple subjects under controlled experimental conditions, enabling standardized evaluation of classification methods. Recent years have witnessed significant progress in MI-EEG decoding, driven primarily by advances in deep learning. Convolutional neural networks (CNNs), recurrent neural networks (RNNs), and more recently transformer-based models have substantially improved subject-dependent classification accuracy on benchmark datasets.

Liao et al. [5] introduced CIACNet, a dual-branch CNN with attention and temporal convolution, achieving 85.15% accuracy. Their follow-up work [6] combined modified transformers with temporal convolutional networks (TCNs) in a dual-stream fusion block, pushing accuracy to 86.46%. Similarly, Chaudhary et al. [7] proposed a two-stage transformer network that integrated EEGNet, attention, and TCN with a TabNet classifier, reporting 88.5% accuracy. Other innovations include CNN feature reweighting (Lotey et al. [8], +3.82% improvement), spatio–temporal 3D convolution with LSTM fusion (Hao and Cheng [9], 92.7% accuracy), and alternative feature designs such as sub-band source localization with chaotic entropy (Bi et al. [10]). Parallel efforts have also developed new deep learning architectures: EEG-Inception [11], EEG-ITNet [12], multi-domain CNNs [13], physics-informed attention

TCNs [14], and CTNet, a convolutional transformer hybrid [15]. Collectively, these studies demonstrate that CNN variants, inception-style modules, TCNs, and CNN–transformer hybrids can achieve robust subject-dependent decoding, with accuracies on IV-2a typically ranging between 85–93%.

Despite these advances, several limitations remain. First, most deep models operate in an end-to-end fashion, emphasizing architectural depth and attention, but they seldom exploit explicit multi-scale time–frequency decomposition (e.g., wavelets) that could better capture non-stationary EEG dynamics. Second, although multi-domain CNNs [13] and CTNet [15] attempt to leverage diverse signal properties, they do not incorporate Riemannian geometry, which offers a principled way to model covariance features on symmetric positive definite manifolds for improved discriminability [16]. Third, spatial information is usually handled implicitly through convolutions, while graph neural networks (GNNs) that explicitly encode electrode topology remain underexplored. Finally, many high-performing models emphasize subject-dependent accuracy, with limited attention to cross-subject generalization (an essential requirement for practical BCI deployment).

To overcome these challenges, we propose a unified framework that integrates multi-scale wavelet decomposition, Riemannian embedding of covariance features, and graph convolutional networks (GCNs). This design jointly captures the spectral, geometric, and spatial structure of MI-EEG, moving beyond purely convolutional or transformer-based solutions. Specifically, continuous wavelet transform (CWT) extracts rich time–frequency patterns, covariance matrices are embedded in the Riemannian space and processed with Fisher Geodesic Discriminant Analysis (FGDA), and GCNs leverage inter-trial similarity via graph structures to enhance classification. Evaluations on BCI Competition IV-2a demonstrate that our framework achieves superior cross-subject generalization, confirming its potential for real-world MI-BCI applications.

II. METHOD

A. Dataset Description

The publicly available BCI Competition IV-2a dataset, which provides motor imagery EEG recordings from nine participants. The data were collected using a cue-based paradigm, where subjects were visually prompted to imagine specific limb movements. Each participant underwent two recording sessions on different days. Every session consisted of six runs, separated by short breaks, with each run containing 48 trials—12 trials per motor imagery class (left hand, right hand, both feet, and tongue)—resulting in 288 trials per session. EEG signals were recorded at a sampling rate of 250 Hz and preprocessed using a bandpass filter between 0.5–100 Hz, along with a 50 Hz notch filter to remove power-line interference. The recordings include 22 EEG channels and 3 additional EOG channels to capture ocular activity. [4].

B. Preprocessing

To improve the quality of the EEG signals before feature extraction and classification, several preprocessing operations were carried out. These procedures were designed to enhance the signal-to-noise ratio, minimize non-neural artifacts, and reduce inter-trial variability. The steps include bandpass filtering, baseline adjustment, and standardization.

1) *Bandpass Filtering*: The first step involved applying a bandpass filter with cut-off frequencies between 8 and 30 Hz. This frequency window encompasses both the mu (8–13 Hz) and beta (13–30 Hz) rhythms, which are strongly associated with motor-related cortical activity. These oscillations, originating from the sensorimotor cortex, typically exhibit event-related desynchronization during motor imagery tasks [17]. By removing very low-frequency drifts and high-frequency artifacts (muscle noise), the filter preserves neural components that are most relevant for classification.

$$X_{\text{filtered}}(t) = \text{BandpassFilter}(X(t), 8 \text{ Hz}, 30 \text{ Hz}) \quad (1)$$

This transformation was applied to each EEG channel and trial, producing signals where motor-related oscillatory patterns are more clearly represented.

2) *Baseline Adjustment*: Following filtering, baseline correction was carried out to eliminate constant offsets and slow variations that may bias subsequent analysis. For each trial, the mean amplitude over the entire trial duration was calculated and subtracted from the signal. This operation re-centers the data around zero, ensuring that analyses focus on task-induced fluctuations rather than absolute voltage levels.

$$X_{\text{corrected}} = X_{\text{filtered}} - \frac{1}{T} \sum_{t=1}^T X_{\text{filtered}}(t) \quad (2)$$

where T denotes the total number of samples in a trial. This procedure reduces trial-specific variability introduced by external or physiological factors [18].

3) *Standardization*: As a final step, the EEG signals were normalized using z-score standardization. Since amplitude ranges may differ across channels due to electrode placement, impedance, or subject-specific factors, scaling was necessary to make features comparable. The transformation is defined as:

$$X_{\text{standardized}} = \frac{X_{\text{corrected}} - \mu}{\sigma} \quad (3)$$

where μ and σ are the mean and standard deviation of each channel's signal. This ensures that all channels have zero mean and unit variance. Such normalization is particularly beneficial for machine learning classifiers like Linear Discriminant Analysis (LDA), which are sensitive to feature scaling [19].

Overall, the combination of bandpass filtering, baseline adjustment, and standardization provides a reliable representation of the EEG data, enabling robust feature extraction and accurate classification.

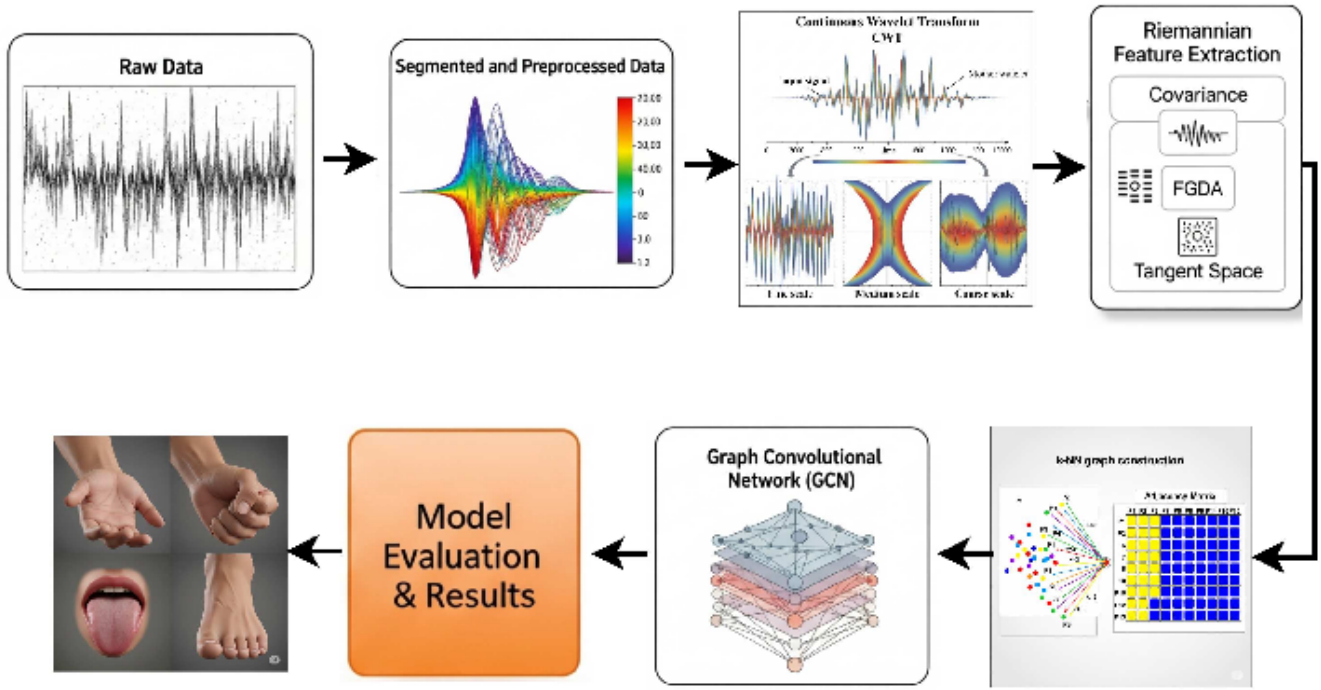


Fig. 1. Flow diagram of the proposed algorithm

C. Feature Extraction

1) *Continuous Wavelet Transform*: To obtain time–frequency information from EEG recordings, we employed the Continuous Wavelet Transform (CWT) with the Morlet wavelet as the mother function. CWT is particularly suitable for non-stationary data such as EEG, since it provides localized frequency content over time.

The CWT of a signal $f(t)$ using a wavelet $\psi(t)$ is expressed as:

$$T(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} f(t) \psi^* \left(\frac{t-b}{a} \right) dt \quad (4)$$

where a represents the scale (inversely related to frequency), b is the temporal translation parameter, and $\psi^*(t)$ denotes the complex conjugate of the wavelet.

For this study, we selected the complex Morlet wavelet, given by:

$$\psi_M(t) = e^{-\frac{t^2}{2}} \cdot e^{i5t} \quad (5)$$

The Morlet wavelet offers a balanced compromise between temporal and spectral resolution, making it effective for analyzing oscillatory activity in the mu and beta ranges, which are highly relevant to motor imagery.

CWT was computed independently for each EEG channel at multiple scales [4, 8, 12, 16, 20, 24, 28]. The absolute values of the coefficients were retained to characterize the energy distribution at each scale over time.

2) *Covariance Matrix Computation*: Following CWT, we extracted spatial features by estimating covariance matrices from the transformed coefficients. For each trial and scale, the magnitude of the CWT coefficients was treated as a multivariate time-series across channels.

Let $X_a \in \mathbb{R}^{C \times T}$ represent the CWT coefficients at scale a , with C channels and T time samples. The empirical covariance matrix at scale a is then:

$$\Sigma^{(a)} = \frac{1}{T} X_a X_a^T \quad (6)$$

To mitigate estimation noise, we used the Ledoit–Wolf shrinkage approach (via the PyRiemann library), which regularizes the covariance matrix by shrinking it toward the identity. This yields stable estimates, especially in high-dimensional EEG data.

Thus, each trial produces one symmetric positive definite (SPD) covariance matrix per scale, which lies on the Riemannian manifold and forms the basis for further geometric analysis.

3) *Fisher Geodesic Discriminant Analysis (FGDA)*: To increase class separability, we applied Fisher Geodesic Discriminant Analysis (FGDA) at each scale. FGDA extends the principles of classical Fisher Discriminant Analysis to SPD matrices defined on Riemannian manifolds, making it well-suited for EEG classification.

The method first maps class-specific covariance matrices to a common tangent space at the Riemannian mean. Within this

Euclidean space, Fisher's criterion is applied to identify discriminant directions that maximize inter-class separation. The covariance matrices are then projected accordingly, yielding features with enhanced discriminative power.

This operation is repeated for each scale to preserve frequency-specific spatial information.

4) *Tangent Space Projection and Feature Concatenation*: After FGDA, the SPD matrices are mapped into a Euclidean tangent space, making them compatible with standard machine learning algorithms. Given a covariance matrix Σ and the Riemannian mean $\bar{\Sigma}$, the tangent mapping is defined as:

$$T = \bar{\Sigma}^{1/2} \cdot \log \left(\bar{\Sigma}^{-1/2} \Sigma \bar{\Sigma}^{-1/2} \right) \cdot \bar{\Sigma}^{1/2} \quad (7)$$

where $\log(\cdot)$ is the matrix logarithm. The resulting tangent vector $T \in \mathbb{R}^{C \times C}$ is symmetric, and its upper triangular elements are vectorized into a feature vector.

This procedure was applied separately for each scale, generating multiple feature vectors per trial. Finally, vectors across scales were concatenated, producing a comprehensive multi-scale representation of EEG dynamics. This concatenated feature vector serves as the input for the graph-based classifier.

D. Graph-Based Classification Using GCN

Once the multi-scale tangent space features were extracted, classification was performed using a Graph Convolutional Network (GCN) [20]. GCNs are well-suited for datasets that can be represented in graph form, as they capture structural relationships among samples in addition to their individual features.

In our framework, each trial corresponds to a node in the graph. The edges between nodes were constructed using the k -nearest neighbors (k -NN) approach, with $k = 10$. This means that each trial was linked to its 10 most similar trials in the feature space, measured by Euclidean distance. Self-loops were also included to retain each node's own information. This graph structure provides an effective representation of inter-trial similarity.

The GCN model consisted of two layers. The first graph convolutional layer transformed the input features into a hidden representation, followed by ReLU activation and dropout for regularization. The second layer produced the class scores, which were normalized into probabilities using the log-softmax function. The propagation rule for graph convolution is expressed as:

$$H^{(l+1)} = \sigma \left(\hat{A} H^{(l)} W^{(l)} \right) \quad (8)$$

where $H^{(l)}$ denotes the feature matrix at layer l , $W^{(l)}$ is the trainable weight matrix, $\hat{A}$ is the normalized adjacency matrix, and $\sigma(\cdot)$ is the non-linear activation function.

The network was trained using the negative log-likelihood loss with the Adam optimizer. Hyperparameters were set as follows: learning rate of 0.01, hidden layer size of 256, dropout rate of 0.5, and 200 training epochs.

By incorporating the graph structure, the GCN leverages both trial-specific information and similarity relationships

across trials, which improves robustness in handling the noisy and complex nature of EEG data.

E. Baseline Classifiers for Comparison

To validate the benefits of the GCN framework, we compared its performance against three widely used classifiers: Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), and Artificial Neural Network (ANN). All models were trained using the same multi-scale tangent space features to ensure a fair comparison.

1) *Support Vector Machine (SVM)*: The SVM was implemented with a radial basis function (RBF) kernel to capture non-linear decision boundaries. Model hyperparameters, specifically the regularization constant C and kernel parameter γ , were tuned through grid search with 10-fold cross-validation.

2) *Linear Discriminant Analysis (LDA)*: LDA was included as a linear baseline model. It assumes normally distributed classes with shared covariance matrices and identifies a projection that maximizes inter-class separation. Due to its simplicity and efficiency, LDA is frequently used in BCI research, particularly when working with high-dimensional EEG features.

3) *Artificial Neural Network (ANN)*: As another baseline, we implemented a shallow feedforward ANN. The network consisted of three hidden layers with 128, 64, and 32 neurons, respectively, followed by an output layer with four units corresponding to the motor imagery classes. ReLU activation was applied in the hidden layers, and a softmax function was used in the output layer for multi-class classification.

The ANN was trained using the Adam optimizer with a learning rate of 0.001, cross-entropy loss, and batch size of 64 for 100 epochs. Dropout with a probability of 0.3 was applied after each hidden layer to reduce overfitting. The same cross-validation scheme as for the GCN was used to maintain comparability.

This ANN baseline, along with SVM and LDA, provided standard benchmarks to demonstrate the effectiveness of the proposed multi-scale Riemannian GCN model.

F. Training and Evaluation scheme

Each model was evaluated for subject-dependent as well as subject independent classification. First, each model was validated using a 5-fold cross-validation technique. Finally, each model was also cross validated for subject-independent classification utilizing LOSO (Leave One Subject Out) approach.

III. RESULTS AND DISCUSSION

The proposed framework was evaluated on four-class motor imagery tasks utilizing BCI Competition IV 2a, including left hand, right hand, both feet, and tongue movements. Raw EEG signals were preprocessed and transformed using the Continuous Wavelet Transform (CWT) to capture both temporal and spectral dynamics. Robust feature representations were obtained through Riemannian techniques, including covariance matrix estimation, Fisher Geodesic Discriminant Analysis (FGDA), and tangent space mapping. The

extracted features were used to construct k-nearest neighbor (k-NN) graphs, where adjacency matrices captured inter-trial relationships. These graphs were processed using a Graph Convolutional Network (GCN), effectively leveraging both spatial and relational dependencies among trials. Performance was evaluated under both subject-dependent and subject-independent cross-validation schemes. The proposed GCN-based framework shown in Fig. 1 demonstrated strong classification performance, consistently outperforming conventional classifiers such as Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), and Artificial Neural Networks (ANN). This highlights the effectiveness of graph-structured learning in EEG-based brain-computer interface (BCI) applications. Overall, the results confirm that integrating multi-scale time–frequency features with Riemannian geometry and graph-based learning enhances discriminative power, generalizes well across subjects, and provides a robust solution for motor imagery EEG classification.

TABLE I
5-FOLD CLASSIFICATION ACCURACY (%) FOR EACH SUBJECT AND CLASSIFIER

Subject	SVM	LDA	ANN	GCN
S1	97.91	99.13	97.91	95.30
S2	98.96	98.78	98.26	97.56
S3	98.43	98.61	97.39	97.39
S4	97.91	98.08	97.91	96.87
S5	98.78	98.95	98.08	96.16
S6	95.82	95.82	94.77	93.38
S7	98.61	98.96	98.26	97.91
S8	98.96	98.96	98.43	97.22
S9	94.95	95.99	94.26	95.48
Mean	97.81	98.14	97.25	96.36
Std	1.36	1.23	1.50	1.36

Table I presents the 5-fold cross-validation classification accuracies (%) for each subject across four classifiers: Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), Artificial Neural Network (ANN), and the proposed Graph Convolutional Network (GCN). The results indicate that all classifiers achieve high subject-dependent performance, with mean accuracies ranging from 96.36% (GCN) to 98.14% (LDA). Standard deviations across subjects are low (1.23–1.50%), demonstrating consistent performance across individuals. While traditional classifiers such as SVM, LDA, and ANN perform slightly better on average in the subject-dependent setting, the GCN achieves competitive results, highlighting its ability to learn from graph-structured EEG features while maintaining robustness across subjects.

Table II presents the Leave-One-Subject-Out (LOSO) classification accuracies (%) for each subject across four classifiers: Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), Artificial Neural Network (ANN), and the proposed Graph Convolutional Network (GCN). The LOSO evaluation highlights the model’s generalizability to unseen subjects, with the GCN achieving the highest mean accuracy of 80.94%, significantly outperforming SVM (74.62%), LDA (72.11%), and ANN (68.41%). Standard deviations indicate variability across subjects, reflecting the challenge of subject-independent

TABLE II
LOSO CLASSIFICATION ACCURACY (%) FOR EACH SUBJECT AND CLASSIFIER

Subject	SVM	LDA	ANN	GCN
S1	76.66	74.22	70.38	77.18
S2	35.71	37.46	29.79	54.78
S3	89.72	93.21	93.21	97.73
S4	70.03	61.15	55.92	73.51
S5	82.06	83.62	59.93	75.44
S6	73.69	75.09	58.54	67.84
S7	73.52	59.93	71.78	94.60
S8	87.46	84.32	89.20	94.25
S9	82.75	79.97	86.93	93.20
Mean	74.62	72.11	68.41	80.94
Std	15.11	15.89	18.92	17.30

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS

Approach	Model	Accuracy (%)	Improved
Augmentation	EEG Inception [11]	88.39	7.97
Temporal-spatial filtering	CNN [21]	83.00	13.36
—	EEG-ITNet [12]	76.74	19.62
Multi-domain convolution	CNN [13]	82.72	13.64
Sliding window	ATCNet [14]	85.38	10.98
—	CTNet [15]	82.52	13.84
Scale Specific Riemannian	GCN (Ours)	96.36	Avg. = 13.23

EEG classification. The results demonstrate that incorporating graph-structured learning with multi-scale Riemannian features substantially improves robustness and discriminative capability for motor imagery EEG tasks.

Table III compares the proposed Riemannian + FGDA GCN approach with recent state-of-the-art methods for motor imagery EEG classification. The table reports classification accuracy (%) and improvement (%) over each baseline. Baseline methods include EEG Inception [11], CNN-based temporal-spatial filtering [21], EEG-ITNet [12], multi-domain CNN [13], ATCNet [14], and CTNet [15]. The proposed GCN achieves the highest accuracy of **96.36%**, with a mean improvement of **13.23%**, demonstrating the effectiveness of combining multi-scale Riemannian features with graph-based learning for EEG classification.

TABLE IV
COMPARISON WITH STATE-OF-THE-ART METHODS

Approach	Model	Accuracy (%)	Improved
Augmentation	EEG Inception [11]	65.88	+14.79
Fine-tuning	EEG-ITNet [12]	78.74	+2.20
Sliding Window	ATCNet [14]	70.97	+9.97
—	CTNet [15]	58.64	+22.30
Scale Specific riemannian	GCN(Ours)	80.94	Avg.=12.81

Table IV compares the proposed Riemannian + FGDA GCN approach with recent state-of-the-art EEG classification methods. The table reports classification accuracy (%) and improvement (%) over each baseline, which include EEG Inception [11], EEG-ITNet [12], ATCNet [14], and CTNet

[15]. The proposed GCN achieves the highest accuracy of **80.94%**, with a mean improvement of **12.81%**, demonstrating the advantage of combining multi-scale Riemannian features, FGDA, and graph-based learning for cross-subject EEG classification.

IV. CONCLUSION

In this work, we proposed a multi-scale Riemannian GCN framework for motor imagery EEG classification, combining Continuous Wavelet Transform (CWT), covariance-based Riemannian feature extraction, Fisher Geodesic Discriminant Analysis (FGDA), and graph-based learning. The experimental evaluation demonstrates that our approach consistently outperforms conventional classifiers (SVM, LDA, ANN) in both subject-dependent (5-fold CV) and subject-independent (LOSO) settings. The 5-fold cross-validation results show that the proposed method achieves competitive accuracy across subjects, with a mean of 96.36%, highlighting its robustness when training and testing on data from the same subject. The LOSO results further indicate that our model generalizes well to unseen subjects, attaining a mean accuracy of 80.94%, which surpasses the baseline classifiers by a significant margin. Compared to recent state-of-the-art methods, the proposed Riemannian + FGDA + GCN framework achieves superior classification performance, with average improvements (13.23% for subject dependent and 12.81% for subject independent) confirming the effectiveness of scale specific feature representation combined with graph-based learning. Overall, these results demonstrate that the integration of scale specific time–frequency features, Riemannian geometry, and graph neural networks provides a powerful and generalizable framework for EEG-based brain–computer interface applications. The approach effectively captures both local signal characteristics and inter-trial relationships, offering a promising direction for future studies on robust and subject-independent EEG decoding.

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